

4.5 Probability generating functions

Further Probability and Statistics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A probability generating function $G_X(t) = E(t^X) = \sum p_r t^r$ for non-negative integer X . $G(1)=1$, $G'(1)=E(X)$, and $G''(1)+G'(1)-G'(1)^2 = \text{Var}(X)$. Independent sums multiply PGFs, allowing convolution without listing every outcome.

Worked example: follow the reasoning

For Bernoulli(p), $G(t) = 1 - p + pt$; two independent Bernoulli variables have PGF $(1 - p + pt)^2$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A variable has $P(X=0)=0.3$ and $P(X=1)=0.7$. Find PGF and $E(X)$ using derivative. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$G(t)=0.3+0.7t$ [1]; $G'(t)=0.7$ [1]; $E(X)=G'(1)=0.7$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.